WORD PROBLEMS

Example 28: There are three poles X, Y and Z. The height of pole Z is $\frac{2}{3}$ of pole Y, the height of pole Y is $\frac{5}{3}$ the pole X. Find the height of pole Z, if the height of pole X is $\frac{99}{25}$ m.

Solution: Height of pole X = $\frac{99}{25}$ m; Height of pole Y = $\frac{5}{3} \times \frac{99}{25}$ m = $\frac{33}{5}$ m

Height of pole Z is $\frac{2}{3}$ of pole Y, i.e., $\frac{2}{3} \times \frac{33}{5}$ m = $\frac{22}{5}$ m

Example 29: The area of a rectangle is $45\frac{5}{16}$ cm^2. If its length is $7\frac{1}{4}$ cm, find the breadth.

Solution: Area of the rectangle = $45\frac{5}{16}$ cm^2 = $\frac{725}{16}$ cm^2; Lenght = $7\frac{1}{4}$ cm = $\frac{29}{4}$ cm

Area of a rectangle = length × breadth $\Rightarrow \frac{725}{16} = \frac{29}{4} \times$ Breadth

$\Rightarrow$ Breadth = $\frac{725}{16} \div \frac{29}{4} = \frac{725}{16} \times \frac{4}{29} = \frac{25}{4} = 6\frac{1}{4}$ cm

EXERCISE 1(F)

1. The product of two numbers is $\frac{-9}{11}$. If one of the numbers is $\frac{-3}{22}$, find the other.
2. By what number should you multiply $\frac{-8}{5}$ so that the product is $\frac{-32}{35}$?
3. Manali bought potatoes at the rate of ₹$10\frac{2}{3}$ per kg on Monday. On Tuesday, the price of potatoes increase by ₹$1\frac{1}{2}$ per kg. What was the cost of potatoes on Tuesday?
4. A price of wood $8\frac{3}{4}$m long is cut into four equal pieces. Find the length of each piece.
5. Mr. Singh has monthly income of ₹7,000. He spends $\frac{11}{30}$ of his income on housing, $\frac{2}{5}$ on food and $\frac{1}{10}$ o entertainment. How much money is left with him?
6. A florist had 400 flowers in his shop. He sold $\frac{1}{10}$ of them in the morning. Out of the remaining flowers, were sold in the evening. $\frac{1}{6}$ of the remaining flowers were withered and could not be sold. Find the tot flowers sold and flowers left.
7. On an annual function, $\frac{63}{20}$ m of cloth is required to make a dress for participants. Out of which, $\frac{3}{20}$ m i wasted. If 220 children participated in the event, how much cloth is used? Also, find the quantity of clot wasted.
8. What is the perimeter of a quadrilateral whose four sides measure are $3\frac{1}{6}$ cm, $2\frac{3}{4}$ cm, $4\frac{5}{12}$ cm and $2\frac{1}{2}$ cm
9. A shopkeeper uses 900 g weight instead of 1 kg. Prachi went to the shopkeeper and asked him to give? $\frac{5}{2}$ kg sugar. Find what fraction of sugar was she cheated of?
10. $\frac{3}{10}$ of a wall was completed on day 1, $\frac{4}{14}$ of the wall was completed on day 2, $\frac{3}{15}$ of the wall was complete on day 3. If the total length of the wall is $513\frac{1}{3}$ m, find:
    a. what part of the wall was constructed on day 4?
    b. how many metres of the wall was constructed on day 4?

Real-life Connect

1. Research the main ingredients used in paracetamol tablets commonly prescribed for fever, along with thei quantity per tablet. Represent these quantities as positive or negative rational numbers.
2. Record the time spent on various daily activities like breakfast, school and lunch. Compare these duration with the total hours in a day and express them as rational numbers in standard form.

Life Skills and Values

Radha donated $\frac{1}{6}$ of the money to the orphanage and $\frac{3}{5}$ of the money to savings account and spends $\frac{6}{8}$ of the money on her expenses. If she has ₹ 24,510 in total, find the amount of money she donated in orphanage. What value is depicted here?

Higher Order Thinking Skills (HOTS) Questions

1. Arrange $\frac{4}{15}, \frac{3}{8}, \frac{5}{6}$ and $\frac{7}{18}$ in ascending order. Express the least number as a per cent of the greatest number.
2. Subtract the additive inverse of $\frac{6}{7}$ from the multiplicative inverse of $\frac{-3}{11} \times \frac{44}{18}$.
3. Find 60 rational numbers between the additive inverse of the sum of $\frac{3}{5}$ and $\frac{1}{4}$ and the product of $\frac{6}{7}$ and $\frac{21}{24}$.
4. What number subtracted from $\frac{16}{5} - \frac{7}{45} + \frac{28}{9}$ will give $\frac{62}{15}$?
5. Simplify:

   a. $1 + \cfrac{1}{1 - \cfrac{1}{1 + \frac{1}{2}}}$

   b. $\cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{1 + \frac{1}{4}}}}$

   c. $3 + \cfrac{3}{2 - \cfrac{2}{3 + \cfrac{3}{2 - \frac{4}{7}}}}$

Rewind Yourself

1. Find the sum.

   a. $\frac{3}{8} + \frac{1}{2} + \frac{-2}{5} + \frac{7}{8} + \frac{-3}{2} + \frac{4}{5}$

   b. $\frac{4}{7} + \frac{-11}{15} + \frac{-8}{9} + \frac{-3}{7} + \frac{14}{15} + \frac{5}{9}$

2. Evaluate:

   a. $\frac{3}{5} \times \left(\frac{7}{12} + \frac{14}{15}\right)$

   b. $\frac{4}{5} \times \frac{7}{11} \times \frac{2}{25} \times 0$

3. State 'True' or 'False'.
   a. The negative of a rational number is a positive rational number.
   b. If x and y are two rational numbers such that $x > y$, then $x - y$ is always a positive rational number.
   c. $-1 \times \frac{a}{b}$ is positive, if $\frac{a}{b}$ is negative.
   d. $-1 \times \left(0 - \frac{a}{b}\right)$ is negative, if $\frac{a}{b}$ is positive.
   e. Multiplication distributes over subtraction of rational numbers.
   f. The reciprocal of a positive rational number is always negative.
4. Verify:

   a. $\frac{-2}{3} + \left(\frac{7}{8} + \frac{-3}{5}\right) = \left(\frac{-2}{3} + \frac{7}{8}\right) + \frac{-3}{5}$

   b. $\frac{-3}{4} + \left(\frac{2}{5} + \frac{-4}{7}\right) = \left(\frac{-3}{4} + \frac{2}{5}\right) + \frac{-4}{7}$

   c. $\frac{-3}{4} \times \left(\frac{4}{5} \times \frac{-7}{8}\right) = \left(\frac{-3}{4} \times \frac{4}{5}\right) \times \frac{-7}{8}$

   d. $\frac{-4}{5} \times \left(\frac{5}{7} + \frac{-8}{9}\right) = \left(\frac{-4}{5} \times \frac{5}{7}\right) + \left(\frac{-4}{5} \times \frac{-8}{9}\right)$
5. Taking $x = \frac{8}{15}$, $y = \frac{2}{5}$ and $z = \frac{4}{10}$, verify that $(x \div y) \times z \neq x \div (y \times z)$.
6. Find two rational numbers between $\frac{1}{5}$ and $\frac{1}{2}$.
7. Find seven rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$.
8. For $x = \frac{-4}{9}$ and $y = \frac{5}{11}$, are $|x - y|$ and $|x| - |y|$ equal? If not, then insert two rational numbers between the two resulting numbers.
9. A train covers $80\frac{4}{5}$ km in 1 hour. How much distance will it cover in $4\frac{3}{4}$ hours?
10. Punit travelled $1\frac{1}{8}$ km from home to school, $\frac{5}{6}$ km from school to skating sink, $\frac{2}{3}$ km from skating sink to

11. Concert tickets usually cost ₹$120\frac{4}{5}$ per person. For students, they are priced at $\frac{1}{4}$ of the normal cost. How much will 6 tickets cost for students?

12. The area of a circle is $\frac{49}{14}$ times its circumference. Find the radius of the circle.

13. The area of a rectangle is $27\frac{1}{5}$ cm². If its breadth is 5 cm, find the perimeter of the rectangle.

14. If $\frac{3}{5}$ cup of water is required to make a cup of tea, find the total quantity of water required for 10 cups of

15. Rahul filled 15 litres of petrol for ₹1,530 in his car. What is the cost of one litre of petrol?

Multiple Choice QUESTIONS

Tick (✓) the correct answer.

1. $\frac{7}{20} + \frac{17}{-45} + \frac{-11}{-30} + \frac{-8}{15}$ is equal to _____.

   a. $\frac{7}{36}$ ☐ b. $\frac{8}{35}$ ☐ c. $\frac{-7}{36}$ ☐ d. $\frac{-35}{180}$

2. $\frac{4}{7} + 0 + \frac{-8}{9} + \frac{-13}{7} + \frac{17}{21} = ?$

   a $\frac{63}{86}$ ☐ b. $\frac{-63}{86}$ ☐ c. $\frac{-86}{63}$ ☐ d. $\frac{-63}{27}$

3. Which of the following is not true?

   a. $\frac{-7}{12} + \frac{-6}{13} = \frac{-6}{13} + \frac{-7}{12}$ ☐ b. $\frac{-5}{-9} + 0 = \frac{5}{9}$

   c. $\frac{9}{16} - \frac{7}{8} = \frac{7}{8} - \frac{9}{16}$ ☐ d. $\frac{9}{14} + \left(\frac{11}{3} + \frac{-2}{5}\right) = \left(\frac{9}{14} + \frac{11}{3}\right) + \frac{-2}{5}$

4. What should be added to $\frac{19}{27}$ to make $\frac{-13}{15}$?

   a. $\frac{-135}{212}$ ☐ b. $\frac{-212}{135}$ ☐ c. $\frac{217}{135}$ ☐ d. $\frac{-219}{135}$

5. The sum of additive inverse and multiplicative inverse of 2 is _____.

   a. $\frac{3}{2}$ ☐ b. $\frac{-3}{2}$ ☐ c. $\frac{1}{2}$ ☐ d. $\frac{-1}{2}$

6. The number which is subtracted from $\frac{27}{13}$ to get $\frac{-3}{7}$ is _____.

   a. $\frac{228}{91}$ ☐ b. $\frac{1}{91}$ ☐ c. $\frac{200}{91}$ ☐ d. $\frac{198}{91}$

7. The rational number which is not lying between $\frac{5}{16}$ and $\frac{1}{2}$ is _____.

   a. $\frac{3}{8}$ ☐ b. $\frac{7}{16}$ ☐ c. $\frac{1}{4}$ ☐ d. $\frac{13}{32}$

8. If $\frac{3}{7} + x + \left(\frac{-8}{21}\right) + \frac{5}{22} = -\frac{125}{462}$, then x is _____.

   a. $\frac{125}{462}$ ☐ b. $\frac{-124}{462}$ ☐ c. $\frac{-252}{462}$ ☐ d. $\frac{124}{462}$

9. Multiplicative inverse of '0' is _____.

   a. $\frac{1}{0}$ ☐ b. 0 ☐ c. does not exist ☐ d. 1

10. The property of rational numbers illustrated by the following expression is _____.

    $\left(\frac{2}{9} + \frac{-5}{8}\right) \times \frac{6}{11} = \left(\frac{2}{9} \times \frac{6}{11}\right) + \left(\frac{-5}{8} \times \frac{6}{11}\right)$

    a. Associativity of addition ☐ b. Associativity of multiplication

    c. Distributivity of multiplication over addition ☐ d. Commutativity of addition